

Data Transmission Technologies: A Review of Security, Reliability, Efficiency, and Emerging Intelligent Communication

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Abstract—Data transmission is a fundamental component of modern communication systems, supporting information exchange across wireless networks, the Internet of Things (IoT), cloud platforms, and specialized communication environments. However, increasing data volumes and diverse application requirements have introduced significant challenges related to security, reliability, latency, bandwidth, energy consumption, and computational complexity. This paper presents a review of data transmission technologies and their evolution from conventional communication approaches to intelligent and resource-aware systems. The review examines secure transmission techniques, including encryption and steganography; wireless and IoT communication; power-line and underwater acoustic transmission; free-space optical and photonic communication; and emerging approaches such as model-based, semantic, and AI-enabled communication. It also discusses integrated sensing and communication (ISAC), THz communication, and three-dimensional photonic integration as promising directions for next-generation networks. The comparative analysis shows that different technologies address specific transmission needs while facing challenges such as communication overhead, environmental interference, device constraints, computational complexity, and deployment limitations. Future systems should therefore balance speed with security, reliability, energy efficiency, scalability, privacy, and application requirements.

Index Terms—Data transmission, wireless communication, Internet of Things, network security, energy efficiency, semantic communication, integrated sensing and communication, intelligent networking.

I. INTRODUCTION

Data transmission is the backbone of today's communication systems. It's what makes it possible for our computers, phones, sensors, servers, and entire networks to talk to one another. In today's rapid-paced digital environment, data exchange has become even more significant and extensive than ever, with technologies such as the Internet of Things (IoT), wireless connectivity, cloud platforms, and smart networking. Given all this data flying around, it's no wonder that ensuring it gets to where it needs to go safely, on time and in good form is a top priority for researchers and engineers. However, data doesn't always behave itself, though. Can be sniffed, hijacked, scrambled, dropped and slowed down by congestion. There is a potential for privacy violation, packets can be corrupted, and delays can make the user experience unpleasant. These issues are particularly severe in IoT and wireless applications where devices tend to be small, inexpensive and have limited processing capacity, memory, bandwidth, and

battery capacity[1]. This paper delves deeper into the body of research that has already been carried out in the field of data transmission and the different methods that have been created to improve the security, reliability, privacy and efficiency of data transmission. We discuss some of the classic techniques such as encryption and steganography, as well as some of the more specialized methods, including wireless protocols, IoT communication options, and even AI-inspired networking hacks. We present these in parallel in order to try and emphasise the strengths and weaknesses of each, and how they might be deployed within the real-world system. The end result here is to gain a more concrete and grounded view on what technologies are already being used to address a host of common data transmission issues and where the future research may have an impact [2]. Taxonomy of reviewed data transmission technologies based on the primary source of transmission gain or constraint is depicted in Figure 1.

II. LITERATURE REVIEW

Elhoseny et al. [1] developed a secure medical data transmission framework for healthcare systems based on Internet of Things technologies that can ensure the protection of sensitive patient information while communicating it through a network. The proposed framework integrates AES and RSA encryption with 2D DWT-based steganography that involves encryption of medical data and its subsequent embedding in medical images to guarantee data confidentiality and information hiding. To assess the effectiveness of the proposed framework, Elhoseny et al. [1] have considered various performance measures including PSNR, MSE, BER, SSIM, structural content, and correlation, obtaining high PSNR values and zero BER, which proves the ability to transmit information securely without degrading the quality of medical images. It can be concluded from this study that the use of cryptographic and steganographic methods together can offer a robust security technique for healthcare IoT communication. Nevertheless, there are certain disadvantages associated with the proposed technique that can serve as further research areas.

Papadimitratos et al. [2] have developed the Secure Message Transmission (SMT) protocol, which guarantees secure and efficient data transmission in Mobile Ad Hoc Networks (MANETs). The suggested solution makes use of multiple paths for transmitting the data from the source to the destination to minimize the effect of possible malicious and selfish

nodes located at some intermediate nodes of the path. The protocol ensures the security of the data transmission process without any need to trust all the nodes lying on the intermediate path. The research proved that the suggested method can guarantee the reliability of data transmission in MANETs. Symoudis et al. [3] analyzed the problem of personal data export and transfer in online services in the context of how users feel about this procedure of data access and transfer. The authors performed two online surveys, where 728 and 817 participants were involved. The participants in the survey requested and checked their real data export. The results demonstrated that looking at one's exported data can have positive effects on privacy attitudes of users, such as privacy control and the usage of privacy-enhancing techniques. At the same time, the participants saw little value in transferring personal data to a new service and were more interested in using their data in various complementary services. It is evident that data transmission and portability not only require certain technical skills but are also influenced by privacy and usability issues of users. Yang et al. [4] have conducted a literature review on different methods of power supply and data transmission through metal barriers to sensors placed inside metallic containers. The researchers examined different electromagnetic couplings, such as inductive, capacitive and magnetic resonance couplings, and ultrasonic couplings, which are used for wireless power and data transmission. As a result of the review, the authors have concluded that the main problem with electromagnetic couplings is that they are applicable only for thin, non-ferromagnetic metal barriers at low data rates owing to shielding and skin effect issues. On the contrary, ultrasonic couplings, especially with piezoelectric transmitters, can be efficient both in terms of power transfer and high data rate through different metal barriers. The advantage of ultrasonic couplings is that they can provide simultaneous power delivery and bi-directional data transmission, but they need better physical coupling of transmitters to metal barriers. Multimodal imaging data integrate information from multiple medical imaging modalities, including Positron Emission Tomography (PET), Computed Tomography (CT), Magnetic Resonance Imaging (MRI), X-rays, and other imaging techniques [5].

Ribeiro et al. [6] evaluates the possibilities of using acoustic data transmission under water for monitoring subsea facilities in deep-water areas. The authors utilized the BELLHOP ray-tracing model for the assessment of the following parameters: transmission loss, multipath, ambient noise, and signal-to-noise ratio (SNR) at various frequencies and types of acoustic modem. It was revealed that the efficiency of the transmission is highly frequency- and distance-dependent. A trade-off is observed among data rate, transmission distance, signal quality, and energy consumption. According to the performed evaluation, the 35 kHz modem has a reasonable balance between data transmission and energy consumption, which makes it appropriate for battery-powered subsea monitoring devices.

N. Li et al. [7] presented a goal-driven semantic communi-

cation scheme for effective video delivery in wireless communications by employing Generative AI and Stable Diffusion for video delivery. This technique focuses on key frame extraction and semantic knowledge extraction from the videos, which reduces the volume of data that is required to be communicated over the wireless channels that are bandwidth limited and noisy. In terms of performance, this model performs significantly better than other models like ADJSCC, Latent-Diff DNSC, DeepWiVe, and DVST based on their PSNR, MSE, and FVD scores. Nevertheless, the use of computationally complex Generative AI models can present difficulties.

Gupta et al. [8] addressed the issue of information transfer over communication networks for estimation and control of linear time-invariant systems remotely. The authors examined communication networks with any topology and unreliable communication links, when packets could be lost during transmission. An information processing technique was developed by the authors in order to allow intermediate network nodes to process and pass information in such a way that a remote estimator would be able to produce the best possible estimate of the state in the minimum mean-square-error sense. In addition, the paper analyzes the stability and performance of a closed loop control system with respect to memory-less and independent packet losses. Nevertheless, it should be noted that this methodology mainly addresses linear systems and stochastic packet losses. Campista et al. [9] have presented a technique to enhance the throughput of data transmission through home electrical wiring using PLC. The authors have developed a technique to enhance the functionality of the HomePlug standard by modifying the MAC sub-layer and developing an enhanced collision avoidance technique. This research proves that a good mechanism for medium access control may be helpful in increasing the efficiency of data transmission through existing electrical wiring networks. However, this research is focused mainly on throughput and collision prevention techniques. Other issues such as security and varying channel conditions need further study. Straub [10] have studied MBDT as a way of minimizing the amount of data necessary for scientific information transfer via long-distance communication channels, especially in interplanetary missions. The suggested method involves comparison of the high-resolution gravitational data gathered by the spacecraft with a model existing either beforehand or transmitted alongside it and subsequent transmission of only the changes needed rather than the whole set. Transmission performance of the method has been examined using earth gravitational model data and compared with raw data transfer and standard ZIP-compression. It resulted in a transmission savings of up to 31.8% in case of no maximum-error thresholding and 97.17% with maximum-error thresholding. This indicates the effectiveness of application-aware data transmission, but it necessitates onboard processing and access to an adequate reference model. L. Li et al. [11] have designed a WSN-based data transmission system for power line monitoring, concentrating on the bottleneck problem that arises around the sink nodes within the long-chain network topology of the transmission lines. Campista et al. [8] have developed a multi-

sink cooperation scheme where neighboring sink nodes from different WSNs cooperate to distribute the network load. Multimode and spatial-multiplexing network architecture has been implemented to differentiate the data streams at various levels to balance the loads of the networks and make the communication resources work more efficiently. The channel adjustment is also used by the scheme to minimize the communication contention among nodes. From simulation, it was found that the multi-sink cooperation scheme could effectively overcome the sink-node funnel problem and get a better quality of service (QoS). However, some open issues were pointed out including the issues of transmission bandwidth, communication range, power supply of the nodes, electromagnetic compatibility, and network security.

Malik et al. [12] introduced the concept of Free-Space Optical (FSO) communications and elaborated on its capabilities as a high-capacity wireless communications medium through which data transmission takes place in open air via optical signals, without any need for physical fiber cables. Malik et al. identified some of the major applications of FSO communications and pointed out the benefits associated with this medium, such as high data rates, low cost of implementation, and lack of need for license to use the spectrum. On the other hand, the data transmission capability of FSO communications suffers greatly from the influence of environmental factors like fog, rain, haze, atmospheric turbulence, and scattering.

Corcoran et al. [15] showed that an ultra-high capacity optical data transmission system can be realized based on a novel concept of soliton-crystal micro-combs used as a single integrated optical source. In particular, their design realizes the data transmission line rate of 44.2 Tb/s with spectral efficiency 10.4 bit/s/Hz under the use of 64-QAM modulation and micro-combs spacing of 48.9 GHz in the C-band of telecommunication spectrum. Transmission distance of 75 km with standard optical fiber was realized in laboratory tests and transmission over existing metro optical fiber link was also demonstrated in this work. The key benefits of their system were provided by the stable operation, efficiency, and non-demanding conditions of soliton-crystal micro-comb generation.

The Genome Sequence Archive (GSA) Family is proposed by Chen et al. [14] in response to the fast-growing amount of genomic and multi-omics data. The proposed archive family is composed of GSA, GSA-Human, and OMIX to provide capabilities for storing, managing, searching, and sharing the data. Specifically, GSA stores raw sequence data, GSA-Human controls and manages access to sensitive human genetic data, and OMIX handles various scientific data types. The platform was created to facilitate accessibility and reusability of the vast biological data with the help of proper security and data management measures. The paper emphasizes the significance of scalable data storage solutions for the efficient transmission and sharing of genomic data between researchers. At the same time, the constantly growing amount of genomic data poses the problems of storage capability, data transmission speed,

data quality assurance, interoperability, and secured access. Daudlin et al. [18] developed a three-dimensional photonic-electronic integration platform to tackle the constraints on power consumption and bandwidth in traditional electrical interchip data transfer. It uses an electronic chip and combines photonic transmitters and receivers through the use of dense 3D links, thereby supporting up to 80 channels in a combined area of just 0.3 mm². The developed platform provided an aggregated data rate of 800 Gb/s and channel density of 5.3 Tb/s/mm², while reaching energy efficiencies of 50 fJ/bit for the transmitter and 70 fJ/bit for the receiver at 10 Gb/s per channel. The authors proved that dense parallel optical channels can noticeably improve communication between separate computing chips, which makes the technology very promising for next-generation AI and high-performance computing. Zhang et al. [19] provided a detailed review of the evolution of Integrated Sensing and Communications (ISAC) technology that involves the merging of sensing and communication functionality within the same wireless system architecture. The authors analyzed the progress of ISAC technology in different frequency spectrums (including RF and optical spectrums), the progress from one cell to multiple cells, and the progress from single-mode sensing to multi-mode sensing. The paper also discussed the role of edge intelligence in facilitating real-time data processing and decision-making with lower latency, as well as the security and privacy issues, and standardization by 3GPP, IEEE, and ITU. It is shown in the paper that ISAC technology is very significant in B5G/6G network due to efficient spectrum and hardware usage, as well as supporting intelligent applications. There are still a number of challenges in ISAC, such as interference management, sensing accuracy, network coordination, security, privacy, and standardization issues.

Table I provides a comparative overview of different research studies that have addressed data transmission from several technological perspectives. The studies cover a wide range of transmission technologies, starting from secure message transmission in mobile ad hoc networks and power-line communication to modern approaches such as IoT data compression, UAV-assisted wireless sensor networks, semantic communication, integrated sensing and communication (ISAC), and generative-AI-based communication. Earlier studies mainly focused on improving the reliability and efficiency of conventional communication networks. For example, Papadimitratos et al. [2] emphasized secure and reliable message transmission in mobile ad hoc networks, while Campista et al. [9] investigated improving throughput through power-line communication. Gupta et al. [8] emphasized on reliable data transmission in the remote estimation and control in the case of packet loss. These studies laid the groundwork for solving reliability, routing, network performance and communication efficiency problems. As wireless and optical technologies have developed, researchers started looking for other ways to transmit data at a faster speed and greater performance. The free space optical communication is reviewed by Malik and Singh [12] where the high bandwidth is highly dependent on the environment like fog, precipitation, turbulence in the

TABLE I: Comparative Analysis of Existing Approaches

Author	Transmission Technology	Main Contribution	Limitation
Papadimitratos et al. [2]	Secure Message Transmission	Secure and reliable data transmission in mobile ad hoc networks using multiple communication paths.	Additional communication and routing overhead.
Campista et al. [9]	Power Line Communication	Improved data transmission throughput over home electrical wiring through an enhanced MAC mechanism.	Performance depends on network load and electrical channel conditions.
Gupta et al. [8]	Networked Estimation and Control	Optimized data transmission for remote estimation and control over unreliable networks with packet losses.	Mainly considers linear systems and stochastic packet-loss models.
Straub [10]	Model-Based Data Transmission	Reduced transmission requirements by sending model updates instead of complete gravitational datasets.	Requires an accurate reference model and additional processing.
Malik and Singhv [12]	Free-Space Optical Communication	Reviewed high-bandwidth wireless optical data transmission and its applications.	Strongly affected by fog, rain, atmospheric turbulence, and line-of-sight conditions.
Yang et al. [4]	Ultrasonic/Electromagnetic Transmission	Reviewed wireless power and data transmission through metal barriers for enclosed sensors.	Metal shielding and transducer coupling limit transmission performance.
L. Li and Zhao [11]	Wireless Sensor Network	Improved power-line monitoring data transmission using multi-sink cooperation and spatial multiplexing.	Limited by bandwidth, transmission distance, power supply, and security.
Shevchuk et al. [13]	IoT Data Compression	Reduced transmitted monitoring data and energy consumption through adaptive filtering and compression.	Effectiveness depends on data characteristics and compression techniques.
Chen et al. [14]	Genomic Data Sharing	Provided scalable storage, management, retrieval, and sharing of large-scale genomic datasets.	Rapid data growth creates storage, transmission, interoperability, and security challenges.
Corcoran et al. [15]	Soliton-Crystal Micro-Combs	Achieved 44.2 Tb/s optical data transmission over standard optical fibre using a single-chip source.	Large-scale integration and practical deployment remain challenging.
N. Li et al. [7]	UAV-Assisted Wireless Sensor Network	Optimized uplink data transmission using collaborative beamforming and multi-objective optimization.	UAV mobility, computational complexity, and changing wireless channels create challenges.
Chaccour et al. [16]	Semantic Communication	Reduced transmitted data by focusing on meaningful information and knowledge rather than raw data.	Semantic interoperability and standardization remain open problems.
Su et al. [17]	Generative-AI Semantic Communication	Used generative AI and semantic information to improve wireless video transmission efficiency.	High computational requirements of generative AI models.
Daudlin et al. [18]	Three-Dimensional Photonic Integration	Enabled ultra-low-energy and high-bandwidth inter-chip data transmission using integrated photonics.	Fabrication, thermal management, and optical coupling remain challenging.
Zhang et al. [19]	Integrated Sensing and Communication	Reviewed the evolution of ISAC and its role in future communication and sensing systems.	Interference, sensing accuracy, security, privacy, and standardization issues.
J. Li et al. [20]	Secure ISAC	Investigated security and privacy mechanisms for integrated sensing and communication in 6G networks.	Secure deployment and privacy protection remain challenging.

atmosphere, and line of sight. In the same way, Chen et al. [14] realized optical transmission at extremely high speed based on soliton-crystal micro-combs, laying the groundwork for the future of communication systems. The advent of IoT has added to the need for data transmission. Data from IoT devices is constantly produced and collected for monitoring and sensor data, with many of these devices being energy constrained, processing constrained, and bandwidth constrained. Shevchuk et al. [15] explored adaptive filtering and data compression to decrease the amount of information that must be sent to reduce energy use. N. Li et al. [7] discussed how to enhance the uplink transmission of wireless sensor networks using UAV assistance, through collaborative beamforming and optimization. The studies highlight that, in line with the progress of modern transmission systems, different aspects of the communication speed are no longer the sole criteria that should be taken into account, but also energy efficiency, device

constraints, network environment and scalability. Elhoseny et al. [1] suggested an AES/RSA encryption and DWT based steganography technique to transmit healthcare data securely. Likewise, in recent research on secure ISAC systems, it was revealed that future 6G networks must support communication and sensing services as well as secure networks, addressing security and privacy. Later studies show that there is a change towards intelligent and semantic data transmission. Chaccour et al. [18] studied semantic communication, which aims to convey the semantic information instead of the entire data. Generative-AI-based semantic communication takes this a step further by leveraging AI models to enhance the efficiency of wireless video transmission. While these can greatly decrease the number of transmissions, some of these methods will need a lot of computational power to be implemented practically. Meanwhile, novel technologies like three-dimensional photonic integration, topological THz

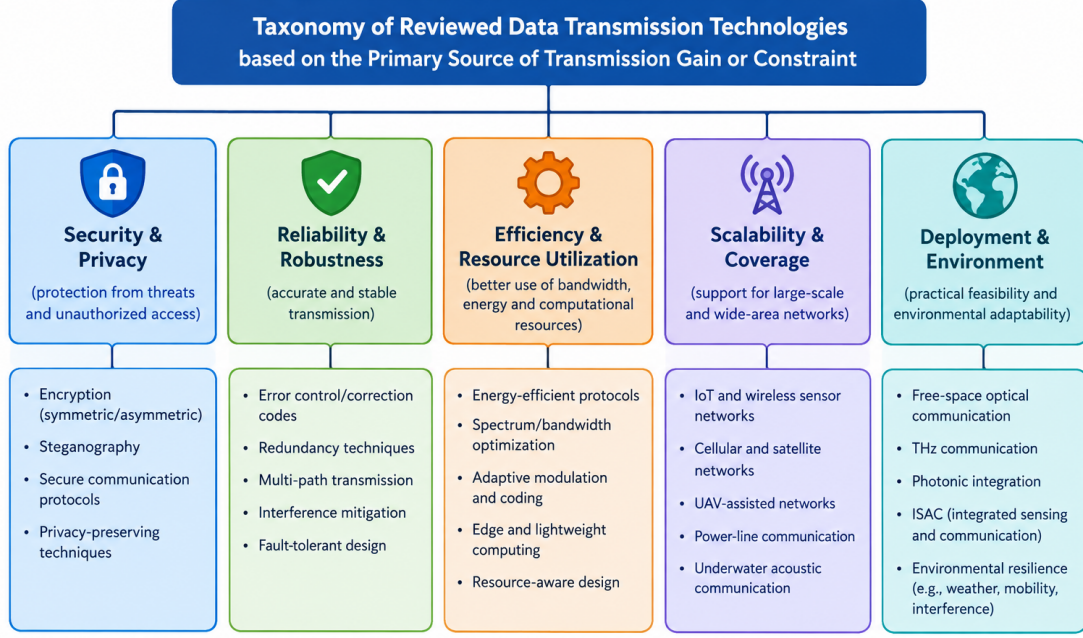


Fig. 1: Taxonomy of reviewed data transmission technologies based on the primary source of transmission gain or constraint.

communication and underwater acoustic communication are providing new opportunities for specialized, long-distance and high-speed data transmission. It can be seen from the table that the field of data transmission research has matured from traditional research areas like reliability and throughput to modern ones like security, energy efficiency, intelligence, scalability, privacy and semantic understanding. While all three strategies have made great strides, each has its own drawbacks: they are computationally intensive, they rely on changing the environment, they require bandwidth that is still limited, they are vulnerable to security threats, they are difficult to fabricate, or they are not well standardized yet. So, any future data transmission system should attempt to achieve a compromise between performance, security, resource usage, and deployability. Su et al. [20] analyzed the security and privacy issues with respect to the Integrated Sensing and Communication (ISAC) for 6G networks and argued that ISAC should not be considered as a technology unless the security threats associated with communication and sensing operations are solved. The study indicated that conventional security methods, such as encryption and key management are not sufficient to address these challenges in 6G network architecture, particularly with high connection density and low latency requirements in 6G network. Thus, the authors stressed that it is crucial to consider security-aware ISAC framework.

The tunable topological directional supercoupler for THz on-chip communications has been proposed and demonstrated by Jia et al. using valley Hall effect in silicon photonic crystals. It is proposed that the device can overcome the limitation of conventional directional coupler in terms of size

and efficiency as it has an extremely small footprint of $0.06 \lambda^2$ in the coupling area and 0.27λ in the coupling length. The tuning range was achieved in this research work by 72.8% over a 24 GHz bandwidth using an all-optical approach. To confirm the capability of communication, a dual-channel on-chip communication system has been demonstrated using the proposed device, which has low insertion loss and could communicate 50 Gbps per channel.

Next generation vision for semantic communication was provided by Chaccour et al. [16], in which information is considered meaningful instead of complete data transmission. Authors suggest a change in the communication architecture from classical data-centric towards knowledge-based and AI-native, in which semantic representation would encode the meaning and significant aspects of information. New performance criteria were proposed in addition to Shannon-based criteria including reasoning capacity. It was shown that semantic communication has great potential in reducing communication needs and increasing the importance of transmitted information in 6G and beyond. Nevertheless, there are many challenges for the future in defining universal semantic representation, evaluation metrics, interoperability between the knowledge of transmitter and receiver, and reasoning-based system deployment.

III. CONCLUSION

In conclusion, the reviewed papers show how rapidly data transmission technologies are evolving towards increasing

bandwidth, minimizing energy usage, ensuring reliability, and improving intelligence of communications. Established technologies such as wireless sensor networks, power-line communication, optical fiber, free-space optics, and underwater acoustic communication are still successfully used in particular environments, while innovative technologies like semantic communication, integrated sensing and communication (ISAC), photonic integration, THz communication, and AI-enabled transmission are responding to the needs of next generation networks. One of the trends observed in the reviewed literature is that of transmitting more meaningful information through minimal usage of communication resources and, thus, minimizing bandwidth, latency, and energy consumption. However, many challenges like those of security, interference from the environment, computational complexity, integration of hardware, scalability, and standardization persist. Hence, the future of data transmission technologies will include AI, photonics, semantics, and secure communication.

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